Name : AKASH SAMPATH

Title : Research_Q_A_Part_2_Programming

Programming Basics : Python :

Variables in Python :

- Variables store values

Code :

```
name = “Akash”
age = 25
salary = 35000

print(name)
print(age)
print(salary)
```

Output :

```
Akash Sampath
25
35000
```

Data Types in Python :

- String : “hello”
- Integer : 10
- Float : 10.5
- Boolean : True/False
- List : [1,2,3]
- Tuple : (1,2,3)

Code :

```
x = 10
y = 3.5
name = “Python”
is_ok = “True

print(type(x))
print(type(y))
print(type(name))
print(type(is_ok))
```

Output :

```
<class ‘int’>
<class ‘float’>
<class ‘str’>
<class ‘bool’>
```

Python Syntax Basics :

- Python uses spaces instead of {}

Example : Code :

```
if 5 > 2:
   print(“5 is greater”)
```

Output :

```
5 is greater
```

Operators in Python :

Arithmetic Operators :

Example : Code :

```
a = 10
b = 3

print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

Output :

```
13
7
30
3.3
```

Comparison Operators :

Example : Code :

```
print(5 > 3)
print(5 ==3)
```

Output :

```
True
False
```

Logical Operators :

Example : Code :

```
print(True and False)
print(True or False)
print(not True)
```

Output :

```
False
True
False
```

Conditional Statements :

IF : Example : Code :

```
age = 18

if age >=18:
    print(“You can vote”)
```

Output : **Input :**

You can vote Enter any age lesser or greater than 18

IF ELSE : Example : Code :

```
num = 5
if num + 2 == 0:
   print(“Even”)
else:
  print(“Odd”)
```

Output :

ODD

IF ELIP ELSE :

Example : Code :

```
marks = 75

if marks >= 90:
    print(“A”)
elif marks >= 60:
    print(“B”)
else:
    print(“C”)
```

Output :

```
B
```

Loops:

FOR Loop : Example : Code :

```
for i in range(1, 6):
    print(i)
```

Output :

1

2

3

4

5

6

WHILE LOOP : Example : Code :

```
i = 1
while i <= 3:
    print(i)
    i += 1
```

Output:

1

2

3

Functions:

- Functions are reusable blocks of code

Example : Code :

```
def add(a, b):
    return a + b

result = add(5, 3)
print(result)
```

Output:

```
8
```

Function without Return : Example code :

```
def greet():
    print(“Hello!”)

greet()
```

Output:

```
Hello!
```

Collections in Python :

List (Changeable) : Example : Code :

```
num = [1,2,3]
nums.append(4)

print(nums)
```

Output:

```
[1, 2, 3, 4]
```

Tuple (Not changeable) : Example : Code :

```
t = (10, 20, 30)
print(t[1])

// 10 + 1 //
```

Output:

```
20
```

Set (Unique Values) : Example : Code:

```
s = {1, 2, 3, 4}
print(s)
```

Output:

{1, 2, 3}

Dictionary : Example code :

```
student = {
    “name”: “Akash”,
    “age”: 25
}

print(student[“name”])
```

Output :

Akash

Input in Python :

- User input is taken using input()

Example : Code :

```
python = input(“Enter your name: “)
print(“Hello” , name)
```

Input :

```
Akash
```

Output :

```
Hello Akash
```

Input Number : Example code :

```
age = int(input(“Enter age: “))
print(age +5)
```

Input :

```
20
```

Output:

```
25
```

Output in Python :

Basic Print : Example : Code :

```
print(“Hello World”)
```

Print Multiple Values : Example code :

```
name = “Akash”
age = 25

print(“Name:”, name, “Age:”, age)
```

Output:

```
Name: Akash Age: 25
```

Formatted Output : Example : Code:

```
name = “Akash”
age = 25

print(f”My name is {name} and I am {age}”)
```

Output:

```
My name is Akash and I am 25
```